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AK Steel 17-4 PH® Precipitation Hardening Stainless Steel, Condition H 900

Categories: [Metal](#); [Ferrous Metal](#); [Martensitic](#); [Stainless Steel](#); [Precipitation Hardening Stainless](#)

Material Notes: AK Steel 17-4 PH® provides an outstanding combination of high strength, good corrosion resistance, good mechanical properties at temperatures up to 600°F (316°C), and good toughness in both base metal and welds. Short-time, low-temperature heat treatments minimize distortion and scaling. This alloy is widely used in the aerospace, chemical, petrochemical, food processing, paper and general metalworking industries.

The material supplied from the mill is in Condition A. After fabrication eight standard heat treatments have been developed to provide a wide range of properties.

Information provided by AK Steel

Key Words: UNS S17400

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Physical Properties	Original Value	Comments
Density	7.80 g/cc	
Mechanical Properties	Original Value	Comments
Hardness, Rockwell C	45	
Tensile Strength, Ultimate	1448 MPa	
Tensile Strength, Yield	1379 MPa @Strain 0.2 %	
Elongation at Break	7.0 %	in 2 inches
Electrical Properties	Original Value	Comments
Electrical Resistivity	0.0000770 ohm-cm	
Thermal Properties	Original Value	Comments
CTE, linear	10.8 µm/m-°C @Temperature 21.0 - 93.0 °C	
	11.7 µm/m-°C @Temperature <=427 °C	
Specific Heat Capacity	0.460 J/g-°C	

Thermal Conductivity 17.9 W/m-K
@Temperature 149 °C

22.6 W/m-K
@Temperature 482 °C

Component Elements Properties	Original Value	Comments
Carbon, C	<= 0.070 %	
Chromium, Cr	15 - 17.5 %	
Copper, Cu	3.0 - 5.0 %	
Iron, Fe	69.91 - 78.85 %	As Remainder
Manganese, Mn	<= 1.0 %	
Nb + Ta	0.15 - 0.45 %	
Nickel, Ni	3.0 - 5.0 %	
Phosphorus, P	<= 0.040 %	
Silicon, Si	<= 1.0 %	
Sulfur, S	<= 0.030 %	

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